



The Palm House, Kew Gardens

Location London, England

Like its immediate predecessor, Joseph Paxton's great glass conservatory at Chatsworth House, the Palm House at Kew Gardens made clear the advantages of prefabrication in iron and glass. It also demonstrated how iron-framed buildings were poised to revolutionize architecture.

The Palm House was built between 1844 and 1848 by the otherwise Classically inclined British architect Decimus Burton and Irish iron engineer Richard Turner. The pair subsequently teamed up to build the Temperate House at Kew. Burton's influence is most obvious in the structure's bulging curves, above all in the high central vault—the whole forming a series of uninterrupted surfaces. The interior, with its slim iron columns, spiral staircases, and elevated walkways, allows itself the occasional decorative flourish.

Library of Sainte-Geneviève

Location Paris, France

Henri Labrouste, the architect of the Library of Sainte-Geneviève, completed in 1850, was among the pioneers of iron-framed construction. The restrained two-story exterior of the library, in effect a kind of Renaissance palazzo with rhythmically repeated arched windows, provides no hint of the technological advances of the vast

▲ THE READING ROOM OF THE LIBRARY OF SAINTE-GENEVIÈVE

reading room within. This room consists of two barrel vaults that run the length of the building, supported by 16 slim cast-iron columns on high stone bases. From them spring elegantly curved transverse iron ribs. The architectural language may have looked backward, but the building's use of iron looked decisively forward. Labrouste's later National Library in Paris would be an even more emphatic assertion of the transforming possibilities offered by industrial technology.